

Optimal Control for Data Science

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Section 1

A Gentle Introduction

Example: Falcon 9 Landing

- ▶ Spectacular example of optimal control.
- ▶ Goal: Safe, precise booster landing.
- ▶ Method: Calculate optimal trajectory & thrust (the "control") to minimize fuel while meeting constraints.

<Placeholder for Falcon 9 landing
video>

*Disclaimer: Educated guess. SpaceX is secretive but known to use the *cvxgen* convex optimization solver by Boyd's group.*

A Toy 2D Model for vertical landing

Goal: Land a 2D "rocket-stick" upright, using minimal fuel.

State

- ▶ $\mathbf{x}(t) = [x, y, \theta, v_x, v_y, \omega]^\top$
position (x, y) , tilt θ , velocities (v_x, v_y) , angular velocity ω
- ▶ $\mathbf{u}(t) = [u_{\text{thrust}}, u_\tau]^\top$
main thrust (push) and body torque (twist)

Cost (minimizes fuel + small regularization)

$$\min_{\mathbf{u}(\cdot), T} J = \int_0^T (|u_{\text{thrust}}(t)| + \lambda u_\tau(t)^2) dt, \quad \lambda \ll 1.$$

Toy Model Cont'd

Dynamics:

$$\dot{x} = v_x \quad (1)$$

$$\dot{y} = v_y \quad (2)$$

$$\dot{\theta} = \omega \quad (3)$$

$$\dot{v}_x = \frac{u_{\text{thrust}}}{m} \sin \theta - \frac{d}{m} v_x \sqrt{v_x^2 + v_y^2} \quad (4)$$

$$\dot{v}_y = \frac{u_{\text{thrust}}}{m} \cos \theta - \frac{d}{m} v_y \sqrt{v_x^2 + v_y^2} - g \quad (5)$$

$$\dot{\omega} = \frac{u_{\tau}}{I} \quad (6)$$

(Acceleration = Gravity + Propulsion + Air Resistance)

Constraints:

Above ground: $y(t) \geq 0$

Thrust limits: $0 \leq u_{\text{thrust}}(t) \leq u_{\text{max}}$

Torque limits: $|u_{\tau}(t)| \leq \tau_{\text{max}}$

Boundary states: $\mathbf{x}(0) = \mathbf{x}_0, \quad \mathbf{x}(T) = \mathbf{x}_f$

Zero terminal velocity: $v_x(T) = v_y(T) = \omega(T) = 0,$

Vertical landing: $\theta(T) = 0$

Simulation Result via Off-the-Shelf Solver (casADi)

<Placeholder for simulation result>

More Familiar Example: Training ResNet

A Residual Network (ResNet) layer:

$$y_{k+1} = y_k + h(y_k, W_k), \quad \text{e.g., } h(y, W) = \text{ReLU}(Wy)$$

In the deep limit, this becomes a controlled ODE:

$$\dot{y}(t) = f(y(t), W(t))$$

Training minimizes a loss $L(y(T))$ by finding optimal weights (controls) $W(t)$.

General Form of Optimal Control

Deterministic: Find control $u(t)$ to minimize

$$J(u) = \Phi(x(T)) + \int_0^T \ell(t, x, u) dt$$

subject to dynamics

$$\dot{x}(t) = f(t, x(t), u(t)), \quad x(0) = x_0$$

and possibly constraints $u(t) \in \mathcal{U}$, $x(t) \in \mathcal{X}$, $g(t, x(t), u(t)) \leq 0$.

Stochastic: Find control process u_t to minimize

$$J(u) = \mathbb{E} \left[\Phi(X_T) + \int_0^T \ell(t, X_t, u_t) dt \right]$$

subject to dynamics

$$dX_t = b(t, X_t, u_t) dt + \sigma(t, X_t) dW_t, \quad X_0 = x_0$$

The Broader Field of Control

- ▶ Control theory is vast; optimal control is one part.
- ▶ Feedback control (e.g., PID) is common for stabilization without a long-term objective.
- ▶ Having an objective function (a goal) provides mathematical structure.
 - ▶ Similar to how linear programming made certain problems tractable.
- ▶ Key Venues: IEEE TAC, Automatica, SICON, CDC, ACC.

Problems of Interest in Optimal Control

- ▶ **Well-posedness:** Does an optimal control exist? Is it unique?
- ▶ **Numerical Algorithms:** How to compute the optimal control?
- ▶ **Stability:** Does the system stay near an equilibrium?
- ▶ **Robustness:** How does the system perform under model uncertainty or disturbances?

Optimal Control vs. Other Fields

- ▶ **Calculus of Variations:** Optimal control is a generalization that handles constraints on controls ($u(t) \in \mathcal{U}$) and states ($x(t) \in \mathcal{X}$).
- ▶ **Functional Optimization:** A specific type where the constraint is a differential equation, providing special structure.
- ▶ **Reinforcement Learning (RL):** In control, the model (dynamics) is typically known. In RL, the model is often learned from data.

Plan for This Talk

Sketch of the talk:

- ▶ Core theoretical ideas of optimal control.
- ▶ Applications in modern data science:
 - ▶ Neural ODEs
 - ▶ Solving High-dimensional PDEs
 - ▶ Generative modeling (diffusion models)

Focus:

- ▶ Ideas and theory, not proofs/algorithms. Like a physics class.

What this talk is NOT about:

- ▶ Engineering algorithms (DDP, iLQR, MPC).
- ▶ Deep connections with Reinforcement Learning.

Section 2

The Maximum Principle

First Approach: Discretize It

The "Discretize then Optimize" (DtO) approach turns the continuous problem into a large nonlinear program.

Discretized Dynamics (Euler method):

$$x_{k+1} = x_k + \Delta t \cdot f(t_k, x_k, u_k), \quad k = 0, \dots, N-1$$

Discretized Objective:

$$\min_{u_0, \dots, u_{N-1}} J \approx \Phi(x_N) + \sum_{k=0}^{N-1} \ell(t_k, x_k, u_k) \Delta t$$

This is a standard numerical approach. But is there a deeper structure?

Can We Go Deeper?

Investigate the Lagrange Multipliers:

$$\mathcal{L} = \Phi(x_N) + \sum_{k=0}^{N-1} \left[\ell_k \Delta t + \lambda_{k+1}^\top (x_k + f_k \Delta t - x_{k+1}) \right]$$

where $\ell_k = \ell(t_k, x_k, u_k)$ and $f_k = f(t_k, x_k, u_k)$.

- What's the relationship between λ_k ?

<Placeholder for whiteboard
derivation>

The Pontryagin's Maximum Principle (PMP)

If (x^*, u^*) is an optimal pair, there exists a *costate* (or *adjoint*) function $\lambda(t)$ such that:

1. **State and Costate Equations:** (x^*, λ) solve the Hamiltonian system:

$$\dot{x}^* = \partial_{\lambda} H(t, x^*, u^*, \lambda), \quad -\dot{\lambda} = \partial_x H(t, x^*, u^*, \lambda)$$

where $H(t, x, u, \lambda) = \ell(t, x, u) + \lambda^\top f(t, x, u)$ is the Hamiltonian.

2. **Minimality Condition:** For a.e. $t \in [0, T]$, $u^*(t)$ minimizes the Hamiltonian:

$$u^*(t) \in \arg \min_{v \in \mathcal{U}} H(t, x^*(t), v, \lambda(t))$$

3. **Transversality Condition:** $\lambda(T) = \nabla \Phi(x^*(T))$.

Remarks on PMP

- ▶ The PMP conditions form a **Two-Point Boundary Value Problem (TPBVP)**:

$$\dot{x}^* = \partial_\lambda H(t, x^*, u^*, \lambda) \quad x^*(0) = x_0 \quad (\text{forward})$$

$$-\dot{\lambda} = \partial_x H(t, x^*, u^*, \lambda) \quad \lambda(T) = \nabla \Phi(x^*(T)) \quad (\text{backward})$$

with $u^*(t) = \arg \min_{v \in \mathcal{U}} H(t, x^*(t), v, \lambda(t))$.

- ▶ It is a **necessary** condition.
- ▶ This inspires "Optimize then Discretize" (OtD) numerical methods:
 - ▶ **Shooting Methods:** Guess $\lambda(0)$, integrate forward, adjust $\lambda(0)$ to satisfy terminal condition on $\lambda(T)$.
 - ▶ **Method of Successive Approximations (MSA):** Iterate: guess control $u_k(t)$, solve for $x(t)$ and $\lambda(t)$, update control to $u_{k+1}(t)$ by minimizing H .

Variational Analysis

Consider a small perturbation around the optimal trajectory (x^*, u^*) :

$$x(t) = x^*(t) + \delta x(t)$$

$$u(t) = u^*(t) + \delta u(t)$$

The first-order change in the objective function J is:

$$\delta J \approx \nabla \Phi(x^*(T))^{\top} \delta x(T) + \int_0^T \left(\frac{\partial \ell}{\partial x}^{\top} \delta x + \frac{\partial \ell}{\partial u}^{\top} \delta u \right) dt$$

Relating δx to δu via the dynamics leads to the PMP.

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derivation>

Application: Neural ODE

Recall the variation formula, the basis for gradient methods:

$$\delta J = \int_0^T \frac{\partial H}{\partial u}(t, x^*, u^*, \lambda)^\top \delta u(t) dt$$

For a **Neural ODE**, we find parameters θ to minimize a loss $L(z(T))$:

$$\frac{dz}{dt} = f(z(t), t; \theta), \quad z(0) = z_0$$

- ▶ **Forward Pass:** Solve ODE from $t = 0$ to T to compute loss $L(z(T))$.
- ▶ **Backward Pass:** How to compute gradient $\frac{dL}{d\theta}$? The adjoint method is an efficient way.

Adjoint Method for Neural ODE

To compute the gradient $\frac{dL}{d\theta}$:

- ▶ **Hamiltonian:** $H = a(t)^\top f(z(t), t; \theta)$ (since running cost $\ell = 0$).
- ▶ The change in loss is $\delta L = \left(\int_0^T \frac{\partial H}{\partial \theta} dt \right)^\top \delta \theta$.
- ▶ So, the gradient is $\frac{dL}{d\theta} = \int_0^T a(t)^\top \frac{\partial f}{\partial \theta}(z(t), t; \theta) dt$.
- ▶ The adjoint state $a(t)$ is found by solving backwards in time:

$$\text{Adjoint Eq.: } \dot{a}(t) = - \left(\frac{\partial f}{\partial z} \right)^\top a(t)$$

$$\text{Terminal Cond.: } a(T) = \frac{\partial L}{\partial z(T)}$$

NN Training as Optimal Control

Advantages:

- ▶ Framework for enforcing constraints (e.g., Lipschitzness).
- ▶ Memory-efficient training via adjoint method (no need to store activations).

Potential Downsides:

- ▶ Slower inference (requires ODE solver).
- ▶ Numerical stability of ODE solver can be an issue.

Literature:

- ▶ Li, Q., Chen, L., Tai, C., & E, W. (2017). Maximum principle based algorithms for deep learning. *JMLR*, 18(165), 1-29.
- ▶ Geshkovski, B., & Zuazua, E. (2022). Turnpike in optimal control of PDEs, ResNets, and beyond. *Acta Numerica*, 31, 135-263.

Section 3

View from Field Aspect

Dynamic Programming

PMP gives an optimal trajectory from a single start point. What about from *any* state (t, x) ?

Define the **Value Function** (or cost-to-go):

$$V(t, x) = \inf_{u_t} \mathbb{E} \left[\Phi(X_T) + \int_t^T \ell(s, X_s, u_s) ds \mid X_t = x \right]$$

Bellman's principle of optimality:

$$V(t, x) = \inf_{u_t} \mathbb{E} \left[\int_t^{t+dt} \ell(s, X_s, u_s) ds + V(t + dt, X_{t+dt}) \right]$$

This leads to a PDE for the value function V .

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derivation>

The Hamilton-Jacobi-Bellman (HJB) Equation

Expanding the dynamic programming principle yields the HJB equation, a PDE for the value function $V(t, x)$.

For the stochastic problem:

$$-\partial_t V = \min_{u \in \mathcal{U}} \left\{ \ell(t, x, u) + (\nabla_x V)^\top b(t, x, u) + \frac{1}{2} \text{Tr}(\sigma \sigma^\top \nabla_x^2 V) \right\}$$

with terminal condition $V(T, x) = \Phi(x)$.

The optimal control is a **feedback** or **closed-loop** policy:

$$u^*(t, x) = \psi(t, x, \nabla_x V(t, x))$$

where ψ is from the minimization in the HJB equation.

Verification Theorem

Theorem (informal): If a C^2 function $W(t, x)$ satisfies:

1. The HJB equation for $t \in [0, T)$,
2. The terminal condition $W(T, x) = \Phi(x)$,

then $W(t, x)$ is the true value function $V(t, x)$, and the resulting policy $u^*(t, x)$ is optimal.

Remarks:

- ▶ **Sufficient** condition.
- ▶ HJB $\implies$ *closed-loop* control $u^*(t, x)$ (a policy for any state).
- ▶ PMP $\implies$ *open-loop* control $u^*(t)$ (a pre-computed plan from a fixed start).
- ▶ The HJB equation is a nonlinear, second-order parabolic PDE.

Viscosity Solutions for HJB

The value function $V(t, x)$ is often not smooth enough for the HJB equation (not C^2).

Viscosity Solution (Idea): A continuous function is a viscosity solution if it satisfies the PDE inequality in a weak sense, tested by smooth functions touching it from above or below.

Why?:

- ▶ This weaker solution concept is exactly what's needed for the value function.
- ▶ Provides a robust framework for existence and uniqueness where classical theory fails.

HJB vs PMP: A Tale of Fields and Particles

Two perspectives for studying dynamical systems:

- ▶ **Field view:** Describes a quantity (e.g., potential) at every point in space-time. Often leads to a PDE.
- ▶ **Particle view:** Tracks trajectories of "particles" or "rays" moving through the field. Often leads to ODEs.

The HJB (field) vs. PMP (particle) duality is a prime example.

A Quick PDE Reminder

- ▶ **Hyperbolic** (e.g., Wave/Advection Eq.): Model conservation laws, transport etc.
- ▶ **Parabolic** (e.g., Heat/Diffusion Eq.): Model diffusion processes, smoothing effects.
- ▶ **Elliptic** (e.g., Laplace Eq.): Model stationary phenomena.

A Simpler Case: Mass Transport in a Fluid

Consider a mass density $C(t, \mathbf{x})$ transported by a fluid with velocity $\mathbf{v}(t, \mathbf{x})$.

Integral Form (Conservation of Mass): Change of mass in a volume Ω equals the net flux across its boundary $\partial\Omega$.

$$\frac{d}{dt} \int_{\Omega} C \, dV = - \oint_{\partial\Omega} (C\mathbf{v}) \cdot d\mathbf{S}$$

Using the divergence theorem (for incompressible fluid, $\nabla \cdot \mathbf{v} = 0$):

$$\int_{\Omega} (\partial_t C + \mathbf{v} \cdot \nabla_{\mathbf{x}} C) \, dV = 0$$

This gives the **advection equation** (a hyperbolic PDE):

$$\partial_t C + \mathbf{v} \cdot \nabla_{\mathbf{x}} C = 0$$

Method of Characteristics for Linear Advection

The advection PDE describes the field. What about a particle in the fluid? Its path $x(t)$ is an ODE called a **characteristic curve**:

$$\dot{x}(t) = v(t, x(t))$$

Along this path, the density C is constant. To see this, take the total derivative of $C(t, x(t))$:

$$\frac{d}{dt}C(t, x(t)) = \underbrace{\partial_t C + v \cdot \nabla_x C}_{=0 \text{ from PDE}} + \nabla_x C \cdot (\dot{x} - v) = 0$$

- ▶ The PDE reduces to a simple ODE ($\frac{dC}{dt} = 0$) along characteristic curves.
- ▶ This is the core idea for solving first-order hyperbolic PDEs.

Method of Characteristics for HJB

- ▶ The deterministic HJB equation ($\sigma = 0$) is a first-order PDE:

$$-\partial_t V = \min_{u \in \mathcal{U}} \left\{ \ell(t, x, u) + (\nabla_x V)^\top f(t, x, u) \right\}$$

- ▶ Recognize a **Hamiltonian-like** structure under minimization.
- ▶ Define the minimized Hamiltonian as
$$H(t, x, p) = \min_u \{ \ell(t, x, u) + p^\top f(t, x, u) \}.$$
(slight abuse of notation here)
- ▶ The HJB equation becomes $\partial_t V + H(t, x, \nabla_x V) = 0$.
- ▶ This is a hyperbolic PDE.
- ▶ Using approach just like the advection equation case, we can derive: (trust me!)

HJB and PMP: The Duality Revealed

- ▶ The characteristic equations of the HJB equation are precisely the state-costate equations from PMP.
 - ▶ $\dot{x}^* = \partial_p H = f(t, x^*, u^*)$ (State equation)
 - ▶ $\dot{p} = -\partial_x H$ (Costate equation)
- ▶ The key connection: The PMP costate $\lambda(t)$ is the gradient of the HJB value function along the optimal path:

$$\lambda(t) = p(t) = \nabla_x V(t, x^*(t))$$

- ▶ Interpretation:
 - ▶ **HJB** defines a "field" of value $V(t, x)$ by propagating terminal cost info backwards in time.
 - ▶ **PMP** finds a single optimal "ray" $x^*(t)$ in this field, driven by the field's gradient, $\lambda(t)$.

Application: Classical Mechanics

The Principle of Least Action: The path of a physical system is an extremum of the *action* integral, S .

$$S = \int_{t_1}^{t_2} L(q, \dot{q}, t) dt$$

where $L = K - U$ is the Lagrangian (Kinetic - Potential Energy).

The necessary condition for this extremum is the Euler-Lagrange equation, which for mechanical systems is equivalent to Newton's second law, $F = ma$. (trust me again!)

Classical Mechanics as an Optimal Control Problem

Frame the Principle of Least Action as an optimal control problem:

- ▶ **Goal:** Find path $q(t)$ that minimizes action S .
- ▶ **State:** Position $q(t)$.
- ▶ **Control:** Velocity $v(t) = \dot{q}(t)$.
- ▶ **Dynamics:** $\dot{q} = v$.
- ▶ **Objective:** Minimize the integral of the Lagrangian.

$$\min_{v(\cdot)} \int_{t_1}^{t_2} L(q(t), v(t), t) dt \quad \text{s.t.} \quad \dot{q} = v, \quad q(t_1) = q_1, \quad q(t_2) = q_2.$$

Applying optimal control machinery to this problem recovers the so-called *Hamiltonian Mechanics* (Fun fact: it was discovered before optimal control!)

Classical Mechanics (Cont'd)

Applying PMP and HJB to the action principle yields Hamiltonian mechanics.

- ▶ **PMP $\implies$ Hamilton's Equations:** The state-costate equations become Hamilton's equations of motion. The costate is momentum ($p = \partial L / \partial \dot{q}$).

$$\dot{q} = \frac{\partial H}{\partial p}, \quad \dot{p} = -\frac{\partial H}{\partial q}$$

- ▶ **HJB $\implies$ Hamilton-Jacobi Equation:** The HJB equation becomes the Hamilton-Jacobi equation for the action S (a first-order, nonlinear PDE).

$$\frac{\partial S}{\partial t} + H\left(q, \frac{\partial S}{\partial q}, t\right) = 0$$

- ▶ Symmetry (Noether's theorem), geometry (Symplectic Structure), etc.

Section 4

BSDE and PDE

From HJB to BSDE

Recall the stochastic HJB equation for the value function $V(t, x)$:

$$-\partial_t V = \min_{u \in \mathcal{U}} \left\{ \ell(t, x, u) + (\nabla_x V)^\top b(t, x, u) \right\} + \frac{1}{2} \text{Tr}(\sigma \sigma^\top \nabla_x^2 V)$$

Assume we have a solution $V(t, x)$. Let's define a process $Y_t = V(t, X_t)$. What are its dynamics?

By Itô's formula, Y_t follows an SDE. This suggests we can represent the PDE solution $V(t, x)$ via a stochastic process.

The Problem:

- ▶ We know the terminal condition $Y_T = V(T, X_T) = \Phi(X_T)$.
- ▶ We do not know the initial condition $Y_0 = V(0, x_0)$.
- ▶ A standard (forward) SDE needs an initial condition. This won't work.

Backward Stochastic Differential Equations (BSDE)

The idea is to define an SDE that runs backward in time.

A BSDE is a pair of processes (Y_t, Z_t) solving:

$$-dY_t = f(t, X_t, Y_t, Z_t)dt - Z_t dW_t, \quad Y_T = \Phi(X_T)$$

In integral form, for $t \in [0, T]$:

$$Y_t = \Phi(X_T) + \int_t^T f(s, X_s, Y_s, Z_s)ds - \int_t^T Z_s dW_s$$

Simple Case ($f = 0$):

- ▶ The equation becomes $Y_t = \Phi(X_T) - \int_t^T Z_s dW_s$.
- ▶ Taking conditional expectation, $Y_t = \mathbb{E}[\Phi(X_T) | \mathcal{F}_t]$.
- ▶ By the Martingale Representation Theorem, there exists a unique process Z_t that represents the martingale Y_t .
- ▶ So, for $f = 0$, both Y_t and Z_t are uniquely determined.

Forward-Backward SDE (FBSDE)

The full system couples a forward SDE for the state with a backward SDE for the value and its gradient.

Uncoupled FBSDE:

$$dX_t = b(t, X_t)dt + \sigma(t, X_t)dW_t, \quad X_0 = x_0 \quad (\text{Forward})$$

$$-dY_t = f(t, X_t, Y_t, Z_t)dt - Z_t dW_t, \quad Y_T = \Phi(X_T) \quad (\text{Backward})$$

This system is the probabilistic representation of a semilinear parabolic PDE.

Remarks:

- ▶ In the general stochastic control problem, the system becomes a **fully coupled FBSDE**, which is the stochastic generalization of PMP.
- ▶ This connection was established by Pardoux & Peng in the early 90s.
- ▶ The solution is a triplet (X_t, Y_t, Z_t) where $Y_t = V(t, X_t)$ and $Z_t = \nabla_x V(t, X_t)\sigma(t, X_t)$.

Solving High-Dimensional PDEs with Deep BSDE

The Challenge: Solving FBSDEs is hard, especially in high dimensions, because we need to find the entire processes (Y_t, Z_t) that satisfy the terminal condition.

The Idea (Deep BSDE Algorithm):

- ▶ Discretize the FBSDE system in time.
- ▶ Use neural networks to approximate the unknown functions at each time step, e.g., $Z_{t_k} \approx \mathcal{NN}_k(X_{t_k}; \theta_k)$.
- ▶ The initial value Y_0 is also a parameter to be learned.
- ▶ Train the networks by minimizing the difference between the computed terminal value Y_T and the true terminal value $\Phi(X_T)$.

This turns solving a high-dimensional PDE into a stochastic optimization problem, solvable with standard deep learning tools.

Literature

- ▶ **Accessible Introduction:**

Perkowski, N. (n.d.). *Backward Stochastic Differential Equations: An Introduction*.

- ▶ **Original BSDE Paper:**

Pardoux, E., & Peng, S. (1990). Adapted solution of a backward stochastic differential equation. *Systems & Control Letters*, 14(1), 55-61.

- ▶ **Deep BSDE Method:**

E, W., Han, J., & Jentzen, A. (2017). Deep Learning-Based Numerical Methods for High-Dimensional Parabolic Partial Differential Equations and Backward Stochastic Differential Equations. *Communications in Mathematics and Statistics*, 5(4), 349-380.

Section 5

Diffusion Optimal Control

Goal: Path Measure Inference

- ▶ A **path measure** is a probability distribution over entire trajectories $\{X_t\}_{t \in [0, T]}$.
- ▶ We start with a simple *prior* process P (e.g., $dX_t = dW_t$).
- ▶ We define a complex *target* measure Q that favors paths with certain properties, via its density:

$$\frac{dQ}{dP} \propto \exp \left\{ \int_0^T f(X_s) ds + g(X_T) \right\}$$

- ▶ **Goal:** Find a simple, controlled process P_u that is "closest" to the target measure Q . We measure closeness by minimizing the KL divergence:

$$\min_u D_{KL}(P_u \| Q)$$

The Variational Inference Approach

The KL minimization problem is a form of variational inference.

- ▶ The core VI identity relates the log-likelihood, KL divergence, and the Evidence Lower Bound (ELBO):

$$\log Z = D_{KL}(P_u \| Q) + \text{ELBO}(u)$$

Minimizing the KL divergence is equivalent to maximizing the ELBO.

- ▶ To express the KL divergence, we need the Radon-Nikodym derivative $\frac{dP_u}{dP}$ for a controlled process $dX_t = \sigma u_t dt + dW_t$.
- ▶ **Girsanov's Theorem** gives us this derivative. The log-density is:

$$\log \left(\frac{dP_u}{dP} \right) = \int_0^T u_s^\top dW_s - \frac{1}{2} \int_0^T \|u_s\|^2 ds$$

From Inference to Control

After substituting the densities into the ELBO and taking expectations (omitting the steps), the problem of maximizing the ELBO becomes:

$$\max_u \mathbb{E}_{P_u} \left[\int_0^T \left(f(X_s) - \frac{1}{2} \|u_s\|^2 \right) ds + g(X_T) \right]$$

The Punchline

The variational inference problem of finding the best path measure approximation is **exactly equivalent** to a stochastic optimal control problem.

A Different perspective: PDE (Hopf-Cole)

The two problems are governed by two different PDEs, linked by a simple transformation.

- ▶ The inference problem (finding Q) is solved by the linear **Feynman-Kac** equation for a function $h(t, x)$.

$$\partial_t h + \mathcal{L}h + f(x)h = 0$$

- ▶ The control problem is solved by the semilinear **HJB** equation for the value function $V(t, x)$.

$$\partial_t V + \sup_u \{ \mathcal{L}_u V + f(x) - \frac{1}{2} \|u\|^2 \} = 0$$

The **Cole-Hopf transformation** $V(t, x) = \log h(t, x)$ converts one PDE into the other.

$$\text{Feynman-Kac} \xrightarrow{\text{Cole-Hopf}} \text{HJB}$$

This provides a powerful analytical link between linear filtering theory and nonlinear optimal control.

Section 6

Conclusion

Take-Home Messages

- ▶ **Flowing Backwards in Time is a Unifying Theme:**
 - ▶ **PMP:** Costate equation integrates backwards from a terminal condition.
 - ▶ **Adjoint Method:** Gradient information flows backwards.
 - ▶ **HJB:** Value function is propagated backwards from terminal cost.
 - ▶ **BSDE:** The defining feature is a backward-in-time stochastic process.
- ▶ **The Same Problem Has Many Faces:**
 - ▶ **Particle vs. Field:** PMP tracks a single optimal path (particle), while HJB defines a value over the whole state space (field).
 - ▶ **Inference vs. Control:** Finding an optimal path measure (variational inference) is equivalent to a stochastic optimal control problem.
 - ▶ These dual perspectives offer different analytical tools and numerical strategies.

Thank You!

Questions?